



No. 13 | Nguyen Thien Minh | CODE 3

All questions will use the same project for easier management (2 DB in the same Project)

Question 1: Initialize the website (2 points)

- Build single database for student management

```
use mystudents;
```

```
go
```

```
INSERT INTO Classes (ClassID, ClassName)
```

```
VALUES
```

```
('C1701', 'SE1701'),
```

```
('C1702', 'SE1702'),
```

```
('C1801', 'SA1801'),
```

```
('C1901', 'SS1901');
```

```
GO
```

```
INSERT INTO Students (StudentID, FullName, ClassID)
```

```
VALUES
```

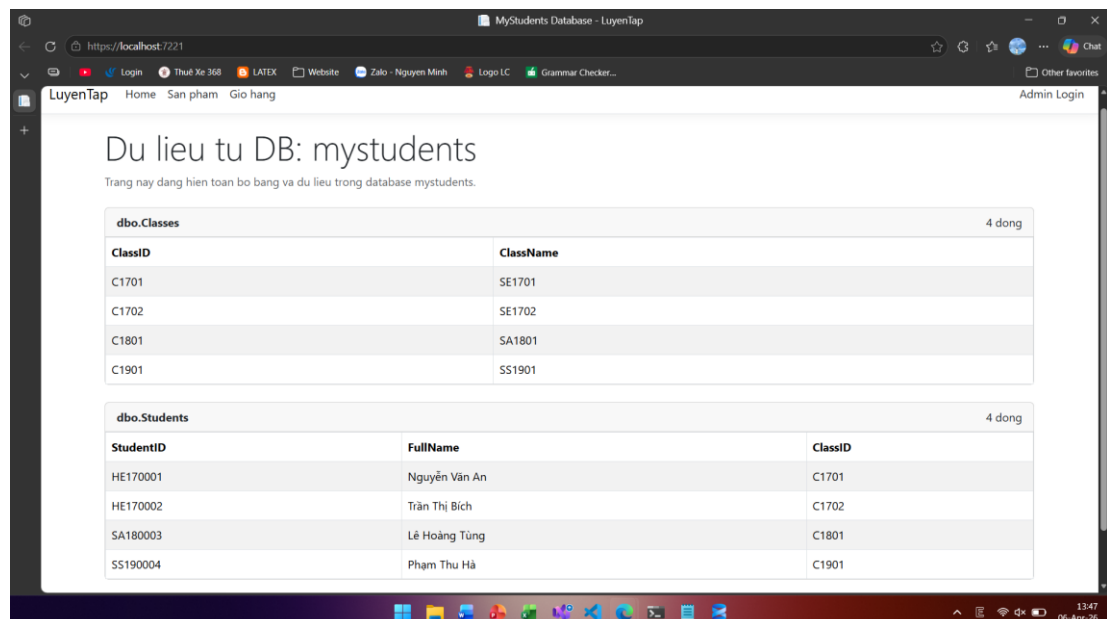
```
('HE170001', N'Nguyễn Văn An', 'C1701'),
```

```
('HE170002', N'Trần Thị Bích', 'C1702'),
```

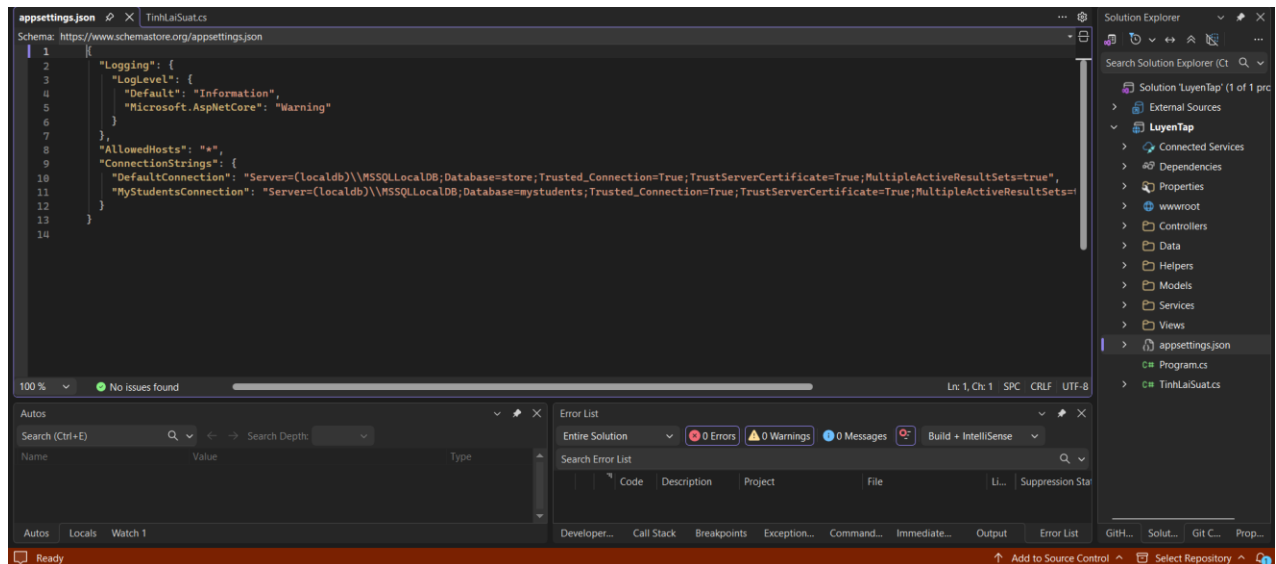
```
('SA180003', N'Lê Hoàng Tùng', 'C1801'),
```

```
('SS190004', N'Phạm Thu Hà', 'C1901');
```

```
GO
```



Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



@model LuyenTap.Models.HomeIndexViewModel

```
@{  
    ViewData["Title"] = "MyStudents Database";  
}
```

```
<div class="mb-4">  
    <h1 class="display-5">Du lieu tu DB: @Model.SourceDatabase</h1>  
    <p class="text-muted">Trang nay dang hien toan bo bang va du lieu trong database  
    mystudents.</p>  
</div>
```

```
@if (!string.IsNullOrEmpty(Model.ErrorMessage))  
{  
    <div class="alert alert-danger">@Model.ErrorMessage</div>  
}  
else if (!Model.Tables.Any())  
{  
    <div class="alert alert-warning">Khong tim thay bang nao trong database  
    mystudents.</div>  
}  
else  
{  
    @foreach (var table in Model.Tables)  
    {  
        <div class="card mb-4">
```



Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

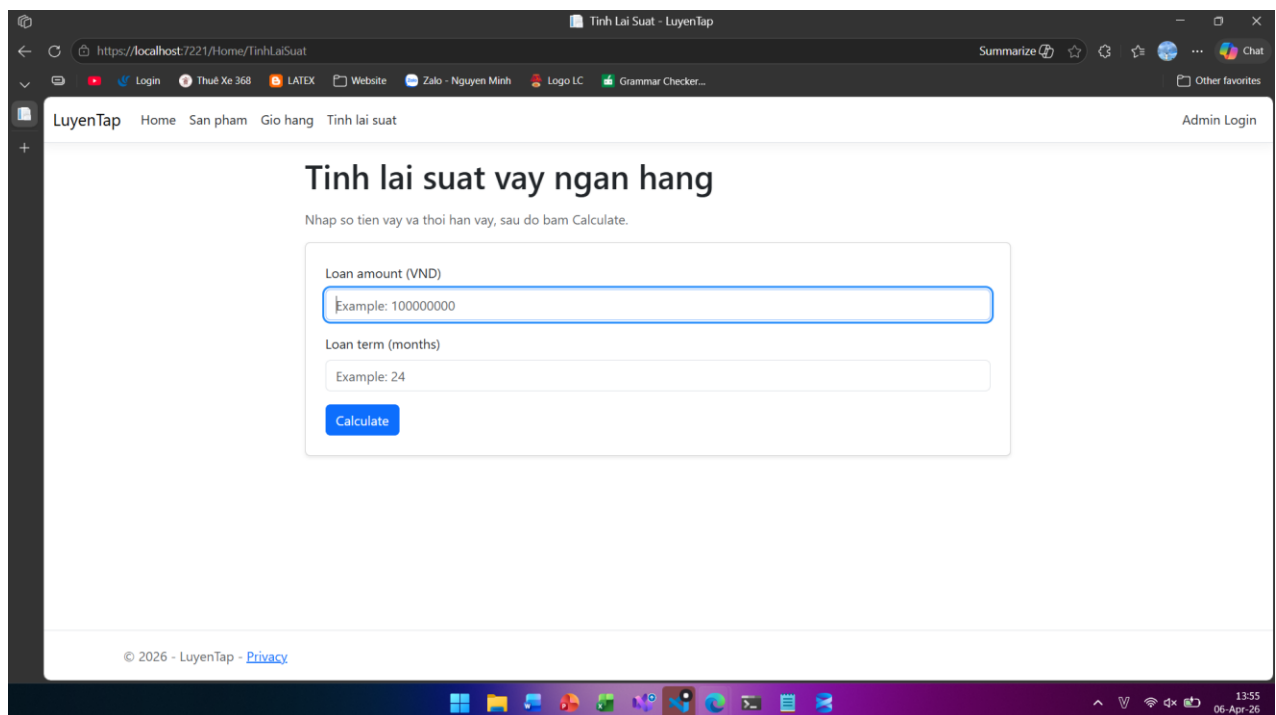
```
<div class="card-header d-flex justify-content-between">
  <strong>@table.Schema.@table.TableName</strong>
  <span>@table.Rows.Count dòng</span>
</div>

<div class="card-body p-0">
  @if (!table.Columns.Any())
  {
    <div class="p-3">Bang không có cột.</div>
  }
  else if (!table.Rows.Any())
  {
    <div class="p-3">Bang không có dữ liệu.</div>
  }
  else
  {
    <div class="table-responsive">
      <table class="table table-striped table-bordered mb-0">
        <thead>
          <tr>
            @foreach (var column in table.Columns)
            {
              <th>@column</th>
            }
          </tr>
        </thead>
        <tbody>
          @foreach (var row in table.Rows)
          {
            <tr>
              @foreach (var column in table.Columns)
              {
                <td>@row[column]</td>
              }
            </tr>
          }
        </tbody>
      </table>
    </div>
  }
</div>
```



```
</div>
}
}
```

Question 2: Create a search function (3 points)



```
@using System.Globalization
```

```
@using LuyenTap
```

```
@{
```

```
    ViewData["Title"] = "Tinh Lai Suat";
```

```
    var amountText = Context.Request.Query["loanAmount"].ToString();
```

```
    var termText = Context.Request.Query["loanTermMonths"].ToString();
```

```
    var hasInput = !string.IsNullOrEmpty(amountText)
```

```
    || !string.IsNullOrEmpty(termText);
```

```
    decimal loanAmount = 0;
```

```
    int loanTermMonths = 0;
```

```
    LoanInterestResult? result = null;
```

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

```
var errorMessage = string.Empty;

if (hasInput)
{
    var amountOk = decimal.TryParse(amountText, NumberStyles.Number,
    CultureInfo.InvariantCulture, out loanAmount)
    || decimal.TryParse(amountText, NumberStyles.Number,
    CultureInfo.CurrentCulture, out loanAmount);

    var termOk = int.TryParse(termText, out loanTermMonths);

    if (!amountOk || !termOk)
    {
        errorMessage = "Invalid input. Please enter numeric values.";
    }
    else
    {
        var calculator = new TinhLaiSuat();
        if (!calculator.TryCalculate(loanAmount, loanTermMonths, out result, out
        errorMessage))
        {
            result = null;
        }
    }
}

<div class="row justify-content-center">
    <div class="col-lg-8">
        <h1 class="mb-3">Tinh lai suat vay ngan hang</h1>
        <p class="text-muted">Nhap so tien vay va thoi han vay, sau do bam Calculate.</p>

        <form method="get" asp-controller="Home" asp-action="TinhLaiSuat" class="card
        p-4 shadow-sm mb-4">
            <div class="mb-3">
                <label for="loanAmount" class="form-label">Loan amount (VND)</label>
                <input id="loanAmount" name="loanAmount" class="form-control"
                value="@amountText" placeholder="Example: 100000000" />
            </div>

            <div class="mb-3">
```



Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

```
<label for="loanTermMonths" class="form-label">Loan term (months)</label>
<input id="loanTermMonths" name="loanTermMonths" class="form-control"
value="@termText" placeholder="Example: 24" />
</div>

<div>
<button type="submit" class="btn btn-primary">Calculate</button>
</div>
</form>
```

```
@if (!string.IsNullOrEmpty(errorMessage))
{
<div class="alert alert-danger">@errorMessage</div>
}

@if (result != null)
{
<div class="card p-4">
<h4 class="mb-3">Result</h4>
<p><strong>Interest rate:</strong> @((result.InterestRate *
100).ToString("0.##"))% / year</p>
<p><strong>Interest amount:</strong> @result.InterestAmount.ToString("N2")
VND</p>
<p class="mb-0"><strong>Total amount payable:</strong>
@result.TotalPayable.ToString("N2") VND</p>
</div>
}
</div>
```

```
</div>
namespace LuyenTap
{
public class LoanInterestResult
{
public decimal InterestRate { get; set; }
public decimal InterestAmount { get; set; }
public decimal TotalPayable { get; set; }
}

public class TinhLaiSuat
{
```



College of Technology
Faculty of Information Technology

Midterm TEST
Web Design and Programming
Mode: Full-time Course:
Location: National Economics University
Duration: 75 minutes

Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

```
public bool TryCalculate(decimal loanAmount, int loanTermMonths, out
LoanInterestResult? result, out string errorMessage)
{
    result = null;
    errorMessage = string.Empty;

    if (loanAmount <= 0)
    {
        errorMessage = "Loan amount must be greater than 0.";
        return false;
    }

    if (loanTermMonths <= 0)
    {
        errorMessage = "Loan term must be greater than 0 month.";
        return false;
    }

    var yearlyRate = GetInterestRateByTerm(loanTermMonths);
    var interestAmount = loanAmount * yearlyRate * loanTermMonths / 12m;

    result = new LoanInterestResult
    {
        InterestRate = yearlyRate,
        InterestAmount = Math.Round(interestAmount, 2),
        TotalPayable = Math.Round(loanAmount + interestAmount, 2)
    };

    return true;
}

private static decimal GetInterestRateByTerm(int loanTermMonths)
{
    if (loanTermMonths <= 12)
    {
        return 0.08m;
    }

    if (loanTermMonths <= 36)
    {
        return 0.10m;
    }
}
```

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

```
    }

    if (loanTermMonths <= 60)
    {
        return 0.12m;
    }

    return 0.135m;
}
}
}

namespace LuyenTap
{
    public class LoanInterestResult
    {
        public decimal InterestRate { get; set; }
        public decimal InterestAmount { get; set; }
        public decimal TotalPayable { get; set; }
    }

    public class TinhLaiSuat
    {
        public bool TryCalculate(decimal loanAmount, int loanTermMonths, out
LoanInterestResult? result, out string errorMessage)
        {
            result = null;
            errorMessage = string.Empty;

            if (loanAmount <= 0)
            {
                errorMessage = "Loan amount must be greater than 0.";
                return false;
            }

            if (loanTermMonths <= 0)
            {
                errorMessage = "Loan term must be greater than 0 month.";
                return false;
            }
        }
    }
}
```




Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

```
var yearlyRate = GetInterestRateByTerm(loanTermMonths);
var interestAmount = loanAmount * yearlyRate * loanTermMonths / 12m;

result = new LoanInterestResult
{
    InterestRate = yearlyRate,
    InterestAmount = Math.Round(interestAmount, 2),
    TotalPayable = Math.Round(loanAmount + interestAmount, 2)
};

return true;
}

private static decimal GetInterestRateByTerm(int loanTermMonths)
{
    if (loanTermMonths <= 12)
    {
        return 0.08m;
    }

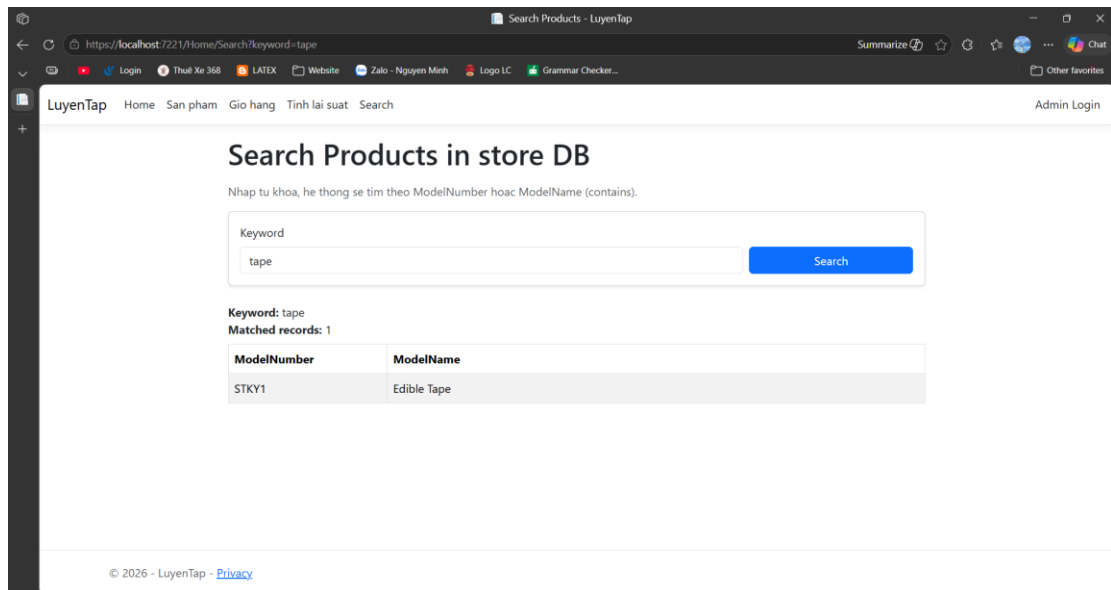
    if (loanTermMonths <= 36)
    {
        return 0.10m;
    }

    if (loanTermMonths <= 60)
    {
        return 0.12m;
    }

    return 0.135m;
}
}
```

Question 3: Create a management function (3 points)

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



Search.cshtml

@model LuyenTap.Models.SearchProductsViewModel

```
@{
```

```
    ViewData["Title"] = "Search Products";
```

```
}
```

```
<div class="row justify-content-center">
```

```
    <div class="col-lg-9">
```

```
        <h1 class="mb-3">Search Products in store DB</h1>
```

```
        <p class="text-muted">Nhập từ khóa, hệ thống sẽ tìm theo ModelNumber  
hoac ModelName (contains).</p>
```

```
        <form method="get" asp-controller="Home" asp-action="Search"  
class="card p-3 shadow-sm mb-4">
```

```
            <div class="row g-2 align-items-end">
```

```
                <div class="col-md-9">
```

```
                    <label for="keyword" class="form-  
label">Keyword</label>
```

```
                    <input id="keyword" name="keyword" class="form-  
control" value="@Model.Keyword" placeholder="Example: pro, 1001, samsung" />
```

```
                </div>
```

```
                <div class="col-md-3 d-grid">
```

```
                    <button type="submit" class="btn btn-  
primary">Search</button>
```

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



```
</div>
</div>
</form>

@if (!string.IsNullOrEmpty(Model.ErrorMessage))
{
    <div class="alert alert-danger">@Model.ErrorMessage</div>
}
else if (!string.IsNullOrEmpty(Model.Keyword))
{
    <div class="mb-2">
        <strong>Keyword:</strong> @Model.Keyword<br />
        <strong>Matched records:</strong> @Model.Results.Count
    </div>

    if (!Model.Results.Any())
    {
        <div class="alert alert-warning">Khong tim thay ban ghi
phu hop.</div>
    }
    else
    {
        <div class="table-responsive">
            <table class="table table-bordered table-striped">
                <thead>
                    <tr>
                        <th>
style="width:220px;">ModelNumber</th>
                        <th>ModelName</th>
                    </tr>
                </thead>
                <tbody>
                    @foreach (var item in Model.Results)
                    {
                        <tr>

                            <td>@item.ModelNumber</td>

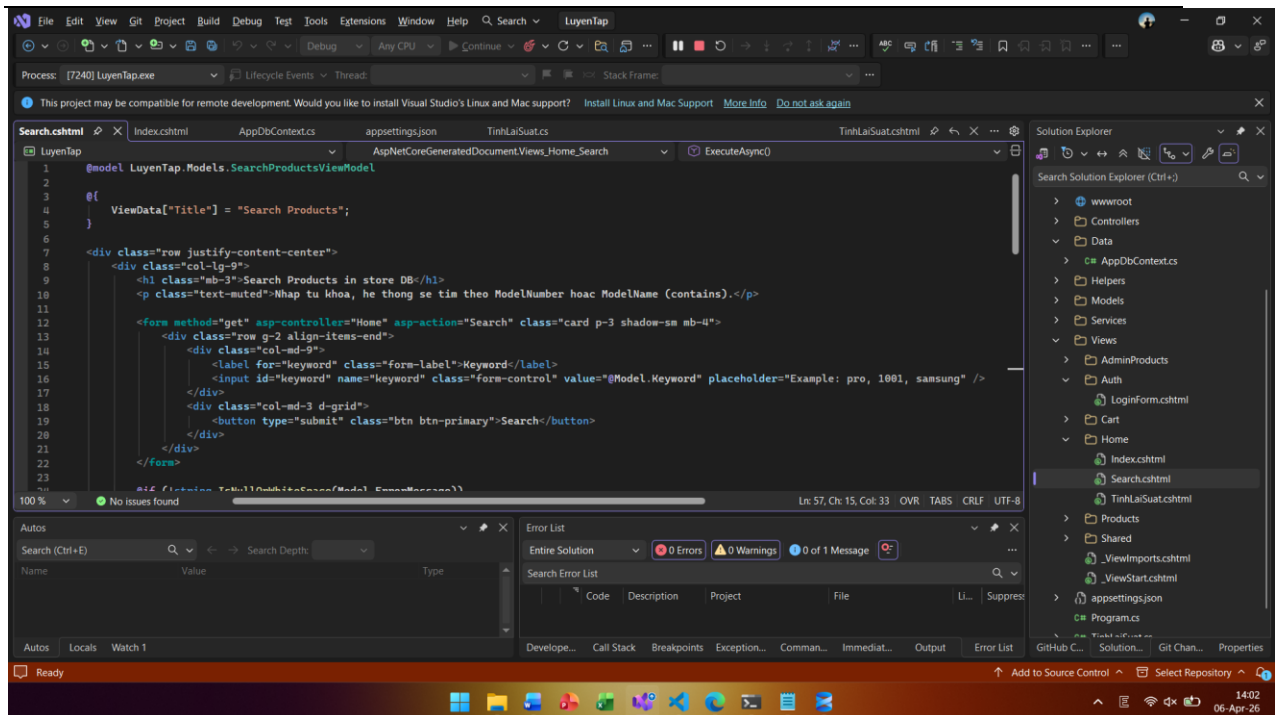
                            <td>@item.ModelName</td>

                        </tr>
                    }
                </tbody>
            </table>
        </div>
    }
}
```

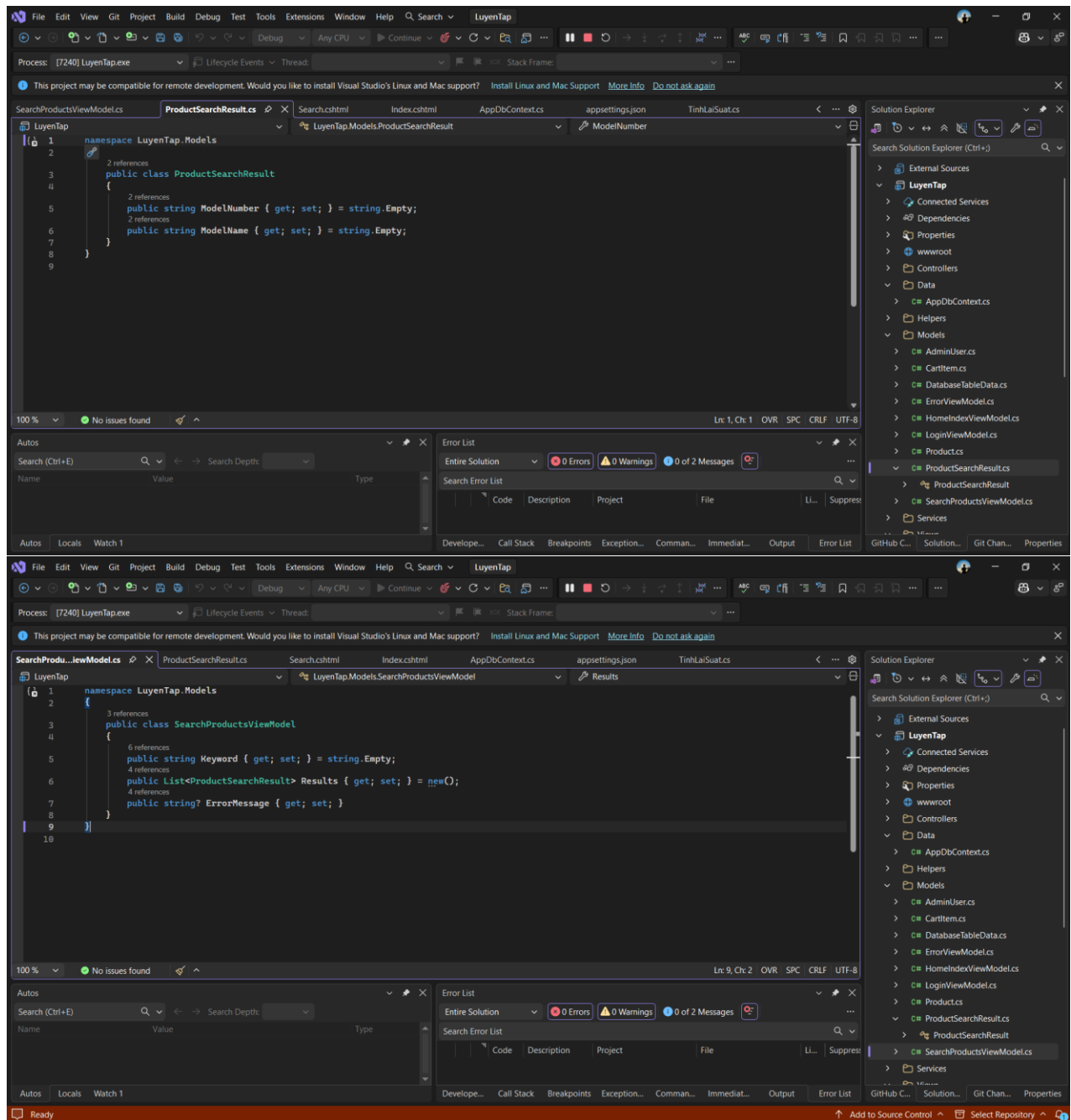
```

</div>
</div>
}
}
</div>
</div>
</table>
</tbody>

```



Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



Question 4: Transaction function (2 points)

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



The screenshot displays a web application interface for product management. The top section, titled "Quan ly san pham", shows a table of products. The bottom section, titled "Sua san pham", shows a form for editing a product.

ID	CategoryID	ModelNumber	ModelName	UnitCost	Thao tac
406	19	GRTWCH9	Multi-Purpose Watch	400 VND	Sua Xoa
404	14	JWLTRANS6	Interpreter Earrings	460 VND	Sua Xoa
402	20	C00LCMB1	Telescoping Comb	400 VND	Sua Xoa
401	14	TCKLR1	Fake Moustache Translator	600 VND	Sua Xoa
400	15	THNKDKE1	Hologram Cufflinks	800 VND	Sua Xoa
399	20	QLT2112	Document Transportation System	300 VND	Sua Xoa
397	14	LSRPTR1	Nonexplosive Cigar	30 VND	Sua Xoa
396	19	BPRECISE00	Dilemma Resolution Device	12 VND	Sua Xoa
394	15	BHONST93	Correction Fluid	2 VND	Sua Xoa
393	20	NOBOOB004U	Speed Bandages	4 VND	Sua Xoa
391	17	BSUR2DUC	Bullet Proof Facial Tissue	80 VND	Sua Xoa

The "Sua san pham" form includes the following fields:

- CategoryId: 19
- ModelNumber: GRTWCH9
- ModelName: Multi-Purpose Watch
- UnitCost: 399.99
- Description: In the tradition of famous spy movies, the Multi Purpose Watch comes with every convenience! Installed with lighter, TV, camera, schedule-organizing software, MP3 player, water purifier, spotlight, and tire pump. Special feature: Displays current date and time. Kitchen sink add-on will be available in the fall of 2001.

Buttons: Cap nhat, Huy

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



```
using System.ComponentModel.DataAnnotations;  
using System.ComponentModel.DataAnnotations.Schema;
```

```
namespace LuyenTap.Models  
{  
    [Table("Products")]  
    public class Product  
    {  
        [Key]  
        [Column("ProductID")]  
        public int Id { get; set; }  
  
        [Column("CategoryID")]  
        [Range(1, int.MaxValue, ErrorMessage = "CategoryID phải lớn hơn 0")]  
        public int CategoryId { get; set; }  
  
        [Column("ModelNumber")]  
        [StringLength(50)]  
        public string? ModelNumber { get; set; }  
    }  
}
```

Ghi chú: Cán bộ coi thi KHÔNG giải thích gì thêm. Sinh viên ĐƯỢC sử dụng tài liệu, laptop, bài thi giống nhau sẽ bị 0 điểm.



College of Technology
Faculty of Information Technology

Midterm TEST
Web Design and Programming
Mode: **Full-time** Course:
Location: **National Economics University**
Duration: 75 minutes

Github Repo: [nguyenminh121/LapTrinhCSharp_at_kiemtra](https://github.com/nguyenminh121/LapTrinhCSharp_at_kiemtra)

```
[Column("ModelName")]
[StringLength(100)]
public string? ModelName { get; set; }

[Column("ProductImage")]
[StringLength(255)]
public string? ProductImage { get; set; }

[Column("UnitCost", TypeName = "money")]
[Range(0.01, 999999999, ErrorMessage = "Unit cost phải lớn hơn 0")]
public decimal UnitCost { get; set; }

[Column("Description")]
[StringLength(1000)]
public string? Description { get; set; }
}
}
```